Intro to Algebra

A Beginner's Guide to Algebraic Thinking

This short course introduces the fundamental concepts of algebra including variables, expressions, order of operations, and solving simple equations. Perfect for students beginning their algebra journey.

Lazard Legacy Math Academy

Empowering Students Through Mathematics

New Orleans, Louisiana

Section 1: What is Algebra? Variables & Expressions

Algebra is a branch of mathematics that uses letters (called **variables**) to represent unknown numbers. It helps us solve problems, find patterns, and describe relationships between quantities.

Why Learn Algebra?

• It builds problem-solving and logical thinking skills

• It's used in science, engineering, business, and everyday life

• It's required for higher math courses and college entrance

• It helps you understand patterns and make predictions

Variables and Constants

A **variable** is a letter that stands for an unknown number (like x, y, n). A **constant** is a number that doesn't change (like 3, 7, –2).

Algebraic Expressions

An **algebraic expression** combines numbers, variables, and operations (+, –, ×, ÷). Examples: $3x + 2$, $5y - 7$, $2a + 3b$

Worked Examples

Example 1: Write an expression: a number plus 8

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Answer: n + 8
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Example 2: Write an expression: four times a number

Answer: $4x$

Example 3: Write an expression: a number divided by 3, minus 2

Answer: $x/3 - 2$

Example 4: Evaluate $5x + 2$ when $x = 3$

Answer: $5(3) + 2 = 15 + 2 = 17$

Example 5: Evaluate $2a - b$ when $a = 4$, $b = 1$

Answer: $2(4) - 1 = 8 - 1 = 7$

Example 6: Identify parts of $7x + 3$: Coefficient = 7, Variable = x, Constant = 3

Example 7: Write: 'six more than twice a number'

Answer: $2n + 6$

Example 8: A pizza costs $12 plus $2 per topping. Expression for t toppings:

Answer: $12 + 2t$

Section 2: Order of Operations (PEMDAS)

When an expression has multiple operations, we must follow a specific order to get the correct answer. This order is remembered by the acronym **PEMDAS**:

Letter	Stands For	Meaning
P	Parentheses	Do operations inside ( ) first
E	Exponents	Calculate powers (^{2}, ^{3}, etc.)
M	Multiplication	Multiply (left to right with D)
D	Division	Divide (left to right with M)
A	Addition	Add (left to right with S)
S	Subtraction	Subtract (left to right with A)

Remember: Multiplication/Division are done together left-to-right (not M before D). Same for Addition/Subtraction.

Worked Examples

Example 1: Simplify: $5 + 3 \times 2$

`= 5 + 6 = 11 (multiply before adding)`

Example 2: Simplify: $8 - 4 \div 2$

`= 8 − 2 = 6 (divide before subtracting)`

Example 3: Simplify: $(3 + 4) \times 2$

`= 7 × 2 = 14 (parentheses first)`

Example 4: Simplify: $2 + 3^2 \times 4$

`= 2 + 9 × 4 = 2 + 36 = 38`

Example 5: Simplify: $20 \div (4 + 1) - 2$

$$= 20 \div 5 - 2 = 4 - 2 = 2$$

Example 6: Simplify: $3(8 - 2)^2 \div 9$

$$= 3(6)^2 \div 9 = 3(36) \div 9 = 108 \div 9 = 12$$

Section 3: Solving Simple Equations

An **equation** says two things are equal. Solving means finding what number the variable equals. Think of it like a balance scale — whatever you do to one side, do to the other.

The Golden Rule: What you do to one side, you MUST do to the other side.

Strategy: Use the OPPOSITE operation

If the equation has...	Undo it with...
Addition (+)	Subtraction (–)
Subtraction (–)	Addition (+)
Multiplication (×)	Division (÷)
Division (÷)	Multiplication (×)

Worked Examples

Example 1: Solve: $x + 5 = 12$

$$x + 5 - 5 = 12 - 5 \rightarrow x = 7$$

Example 2: Solve: $y - 3 = 10$

$$y - 3 + 3 = 10 + 3 \rightarrow y = 13$$

Example 3: Solve: $4n = 20$

$$4n \div 4 = 20 \div 4 \rightarrow n = 5$$

Example 4: Solve: $x/6 = 3$

$$x/6 \times 6 = 3 \times 6 \rightarrow x = 18$$

Example 5: Solve: $2x + 1 = 9$

$$2x = 8 \rightarrow x = 4$$

Example 6: Solve: $3y - 5 = 10$

$$3y = 15 \rightarrow y = 5$$

Example 7: Solve: $x/2 + 4 = 7$

$$x/2 = 3 \rightarrow x = 6$$

Example 8: Solve: $15 = 3 + 4a$

$$12 = 4a \rightarrow a = 3$$

Section 4: Practice Problems

Directions: Solve each problem. Show your work.

Part A: Write Algebraic Expressions (1-10)

1. A number increased by 12

2. Seven less than a number

3. Three times a number plus 4

4. A number divided by 5

5. Twice a number decreased by 6

6. The sum of a number and 15

7. Eight more than four times a number

8. A number cubed minus 10

9. Half of a number plus 3

10. The product of 9 and a number

Part B: Order of Operations (11-20)

11. Simplify: $4 + 6 \times 3$

12. Simplify: $12 \div 3 + 5 \times 2$

13. Simplify: $(8 + 2) \times 5$

14. Simplify: $3^2 + 4 \times 2$

15. Simplify: $20 - 2(3 + 4)$

16. Simplify: $15 \div (1 + 2) + 6$

17. Simplify: $4(5 - 1)^2 \div 8$

18. Simplify: $2^3 + 4^2 - 10$

19. Simplify: $50 - 3(8 - 2)$

20. Simplify: $(6 + 3)^2 \div 3 - 7$

Part C: Solve Equations (21-35)

21. Solve: $x + 7 = 15$

22. Solve: $y - 4 = 9$

23. Solve: $3n = 18$

24. Solve: $x/5 = 4$

25. Solve: $2x + 3 = 11$

26. Solve: $4y - 2 = 14$

27. Solve: $x/3 + 1 = 6$

28. Solve: $5a - 10 = 0$

29. Solve: $6 + 2x = 18$

30. Solve: $3(x + 2) = 15$

31. Solve: $20 = 4y + 8$

32. Solve: $x/4 - 3 = 2$

33. Solve: $7n + 5 = 33$

34. Solve: $2(x - 3) = 10$

35. Solve: $9 = x/2 + 1$

Section 5: Answer Key

Part A: Expressions

1. $n + 12$
2. $n - 7$
3. $3n + 4$
4. $n/5$
5. $2n - 6$
6. $n + 15$
7. $4n + 8$
8. $n^3 - 10$
9. $n/2 + 3$
10. $9n$

Part B: Order of Operations

11. 22
12. 14
13. 50
14. 17
15. 6
16. 11
17. 8
18. 14

19. 32

20. 20

Part C: Equations

21. x = 8

22. y = 13

23. n = 6

24. x = 20

25. x = 4

26. y = 4

27. x = 15

28. a = 2

29. x = 6

30. x = 3

31. y = 3

32. x = 20

33. n = 4

34. x = 8

35. x = 16